

Chapter 8: Testing Set Overfitting & Backtest Validation

Machine Learning for Asset Managers (Marcos M. López de Prado)

AI Finance Seminar

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The Backtest Delusion

- **Historical Fitting vs. Future Alpha:** A backtest is an ex-post historical simulation of monetization, not a proof of causality.
- **Selection Bias on Multiple Testing:** When a quantitative researcher tests N strategy variations and reports only the best result, the backtest suffers from extreme selection bias.
- **Marcos' Second Law:** *"Overfitting is not a feature of an algorithm; it is a feature of a research protocol that fails to track multiple testing."*

Key Chapter Solutions

- **Deflated Sharpe Ratio (DSR):** Corrects the Sharpe ratio for non-normality, track record length, and number of trials N .
- **Probability of Backtest Overfitting (PBO):** Uses Combinatorial Purged Cross-Validation (CPCV) to measure OOS performance decay.
- **Minimum Track Record Length (MinTRL):** Quantifies the minimum years of data required to establish statistical significance.

The Expected Maximum Sharpe Ratio Under $H_0 : \mu_{SR} = 0$

Under the null hypothesis that all N tested strategies have an expected Sharpe ratio of zero, the expected maximum Sharpe ratio $\mathbb{E}[\max_N \{\widehat{SR}_n\}]$ increases with trial count N and variance $V[\widehat{SR}]$ ($\gamma \approx 0.5772$ Euler-Mascheroni constant):

$$\mathbb{E} \left[\max_N \{\widehat{SR}_n\} \right] \approx \sqrt{V[\widehat{SR}]} \left((1 - \gamma)Z^{-1} \left[1 - \frac{1}{N} \right] + \gamma Z^{-1} \left[1 - \frac{1}{Ne} \right] \right)$$

Takeaway: As $N \rightarrow \infty$, you are guaranteed to find an arbitrarily high "in-sample" Sharpe ratio strictly by chance!

Familywise Error Rate (FWER)

Probability of making at least one false discovery across N trials:

$$\text{FWER} = 1 - (1 - \alpha)^N \rightarrow 1.0 \quad \text{as } N \text{ grows}$$

False Discovery Rate (FDR)

Proportion of false positives among all selected strategies. Standard backtesting ignores N , mistaking random noise for true trading edge.

Probabilistic Sharpe Ratio (PSR)

Calculates the probability that an estimated Sharpe ratio $\widehat{SR}$ exceeds a benchmark threshold SR^* , adjusting for non-normal skewness ($\widehat{\gamma}_3$) and kurtosis ($\widehat{\gamma}_4$) over T observations:

$$\text{PSR}(SR^*) = Z \left[\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \widehat{\gamma}_3\widehat{SR} + \frac{\widehat{\gamma}_4-1}{4}\widehat{SR}^2}} \right]$$

Deflated Sharpe Ratio (DSR)

Replaces the static benchmark SR^* with the expected maximum Sharpe ratio under the null hypothesis ($SR_0^* = \mathbb{E}[\max_N\{\widehat{SR}_n\}]$), adjusting for selection bias across N trials and variance $V[\widehat{SR}]$:

$$\text{DSR} = \text{PSR}(SR_0^*) = Z \left[\frac{(\widehat{SR} - SR_0^*)\sqrt{T-1}}{\sqrt{1 - \widehat{\gamma}_3\widehat{SR} + \frac{\widehat{\gamma}_4-1}{4}\widehat{SR}^2}} \right]$$

Combinatorial Purged CV (CPCV)

- Standard k -fold cross-validation generates a single OOS path.
- **CPCV** partitions T observations into N equal groups and selects k groups for testing, generating $\binom{N}{k}$ combinations of In-Sample (IS) and Out-of-Sample (OOS) pairs.
- **Purging & Embargoing**: Removes overlapping labels between IS and OOS sets to prevent lookahead leakage.

Computing PBO

- For each combination c , find optimal IS strategy $n^*(c)$.
- Determine OOS rank r_c^* of strategy $n^*(c)$ among all N strategies, defining relative rank $\omega_c = \frac{r_c^*}{N+1} \in (0, 1)$.
- **PBO Formulation**:

$$\text{PBO} = \int_{-\infty}^0 f(\lambda) d\lambda = P[\omega \leq 0.5]$$

where $\lambda = \ln\left(\frac{\omega}{1-\omega}\right)$ is log-odds rank degradation.

How Much Data Do You Need?

MinTRL calculates the minimum number of years/observations $T_{\min}$ required to reject $H_0 : SR \leq SR^*$ at confidence level α , accounting for fat tails and skewness:

$$\text{MinTRL} = 1 + \left(1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2 \right) \left(\frac{Z_\alpha}{\widehat{SR} - SR^*} \right)^2$$

Practical Numerical Example

- **Daily Strategy:** Annualized $\widehat{SR} = 1.0$ (daily $\widehat{SR} \approx 0.063$), zero skewness ($\hat{\gamma}_3 = 0$), kurtosis $\hat{\gamma}_4 = 6$ (heavy-tailed), testing against $SR^* = 0$.
- **Standard Gaussian Assumption:** Suggests ≈ 2.7 years of daily data is sufficient.
- **MinTRL Correction:** High kurtosis ($\hat{\gamma}_4 = 6$) increases required data to ≥ 5.9 years! Ignoring fat tails leads to premature live deployment.

Performance Degradation Chart

- Plotting IS Sharpe ratio vs. corresponding OOS Sharpe ratio across all CPCV combinations.
- **No Overfitting:** Positive linear slope ($m \approx 1$). High IS performance translates directly to high OOS performance.
- **Severe Overfitting:** Zero or negative slope. The best IS strategies systematically underperform OOS (mean-reverting to noise).

First & Second Order Stochastic Dominance

- Evaluates whether OOS return distribution $F(r)$ stochastically dominates benchmark $G(r)$:

- **1st-Order Stochastic Dominance:**

$$F(r) \leq G(r) \quad \forall r \quad (\text{Wealth Maximization})$$

- **2nd-Order Stochastic Dominance:**

$$\int_{-\infty}^x [G(r) - F(r)] dr \geq 0 \quad \forall x \quad (\text{Risk Aversion})$$

Protocol Requirements

- 1 **Track All Trials (N):** Record every backtest iteration, parameter change, and dataset tweak in a central audit log.
- 2 **Compute DSR & PBO Ex-Ante:** Never present an isolated Sharpe ratio without reporting its DSR p-value and PBO percentage.
- 3 **Apply Purging & Embargoing:** Enforce CPCV cross-validation to eliminate label overlap.

Research Philosophy

- 4 **Focus on Feature Importance:** Research economic mechanisms and variable relevance ex-ante rather than running backtest iterations.
- 5 **Avoid Backtest-Driven Optimization:** Do not iterate on strategy code until a backtest "looks good." If a backtest fails, abandon the theoretical premise instead of tweaking parameters.

Core Principles for Quantitative Asset Managers

- ❶ **Backtests Lie Without N :** Any backtest that does not account for the total number of historical trials N is statistically meaningless.
- ❷ **Deflate Your Sharpe Ratio:** Use DSR to adjust for selection bias, short track record lengths, negative skewness, and fat-tailed kurtosis.
- ❸ **Quantify Overfitting via PBO:** Use CPCV to compute PBO. A strategy with $PBO > 0.50$ is more likely to lose money OOS than to succeed.
- ❹ **Respect MinTRL:** Verify that available historical data meets the Minimum Track Record Length requirement before deploying real capital.

Thank You!

Questions & Discussion

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